

Unit-1

Concept of e-Learning

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Communication Tools

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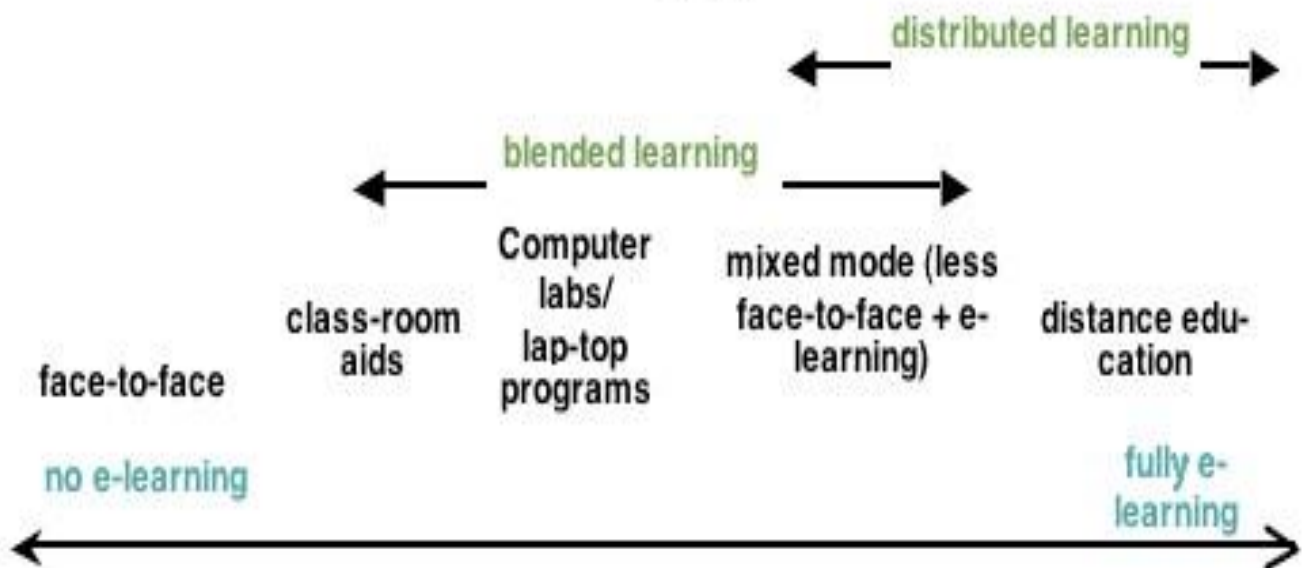
Definition of E-learning

Learning: Learning is the act of acquiring new, or modifying and reinforcing, existing **knowledge, behaviours, skills, values, or preferences** and may involve synthesizing different types of **information**.

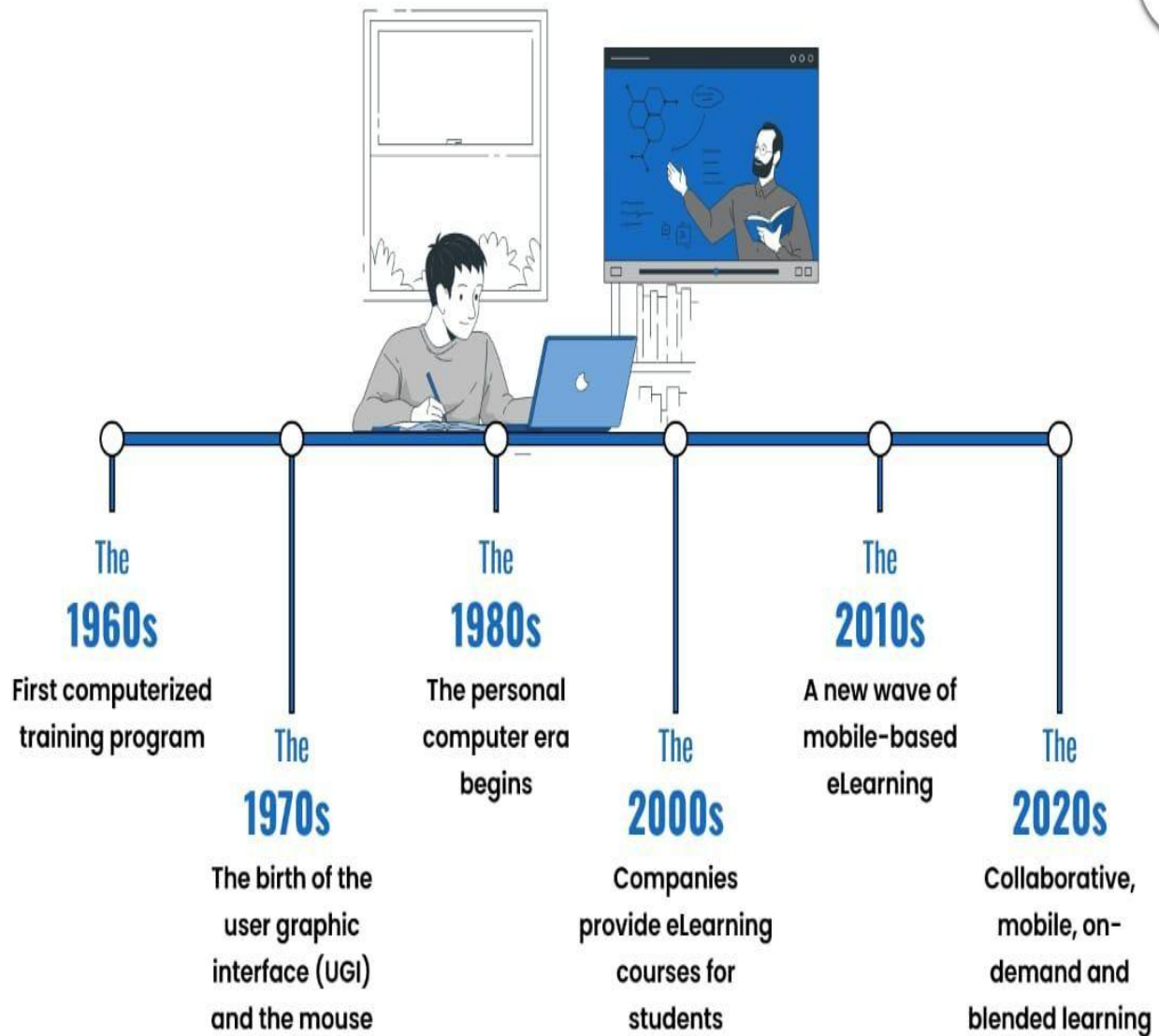
Learning practices in institutions can be either traditional classroom learning with the presence of teachers or it may be a **digital learning** with the use of digitally assisted environment in the classroom, which includes use of digital technologies. **These days traditional learning and digital learning has been transformed to the e-learning where the idea is to conduct teaching learning practices over the online platform like internet.** With e-learning idea is to enforce the virtual learning environments.

Cont..

Figure 1
Different forms of e-learning
(from OECD, 2005 and Bates and Poole,
2003)



A Brief History of eLearning



E-learning:

E-learning is the continuous assimilation of knowledge and skills by adults stimulated by synchronous and asynchronous learning events—and sometimes Knowledge Management outputs—which are authored, delivered, engaged with, supported, and administered using computer and/or internet technologies.

E-learning facilitates education via the [Internet](#), [network](#), or standalone [computer](#). E-learning is essentially the network-enabled transfer of skills and knowledge. E-learning refers to using electronic [applications](#) and processes to learn. E-learning applications and processes include [Web](#)-based learning, computer-based learning, [virtual](#) classrooms and digital collaboration. **Content is delivered via the Internet, intranet/extranet, audio or video tape, satellite TV, and [CD-ROM](#).**

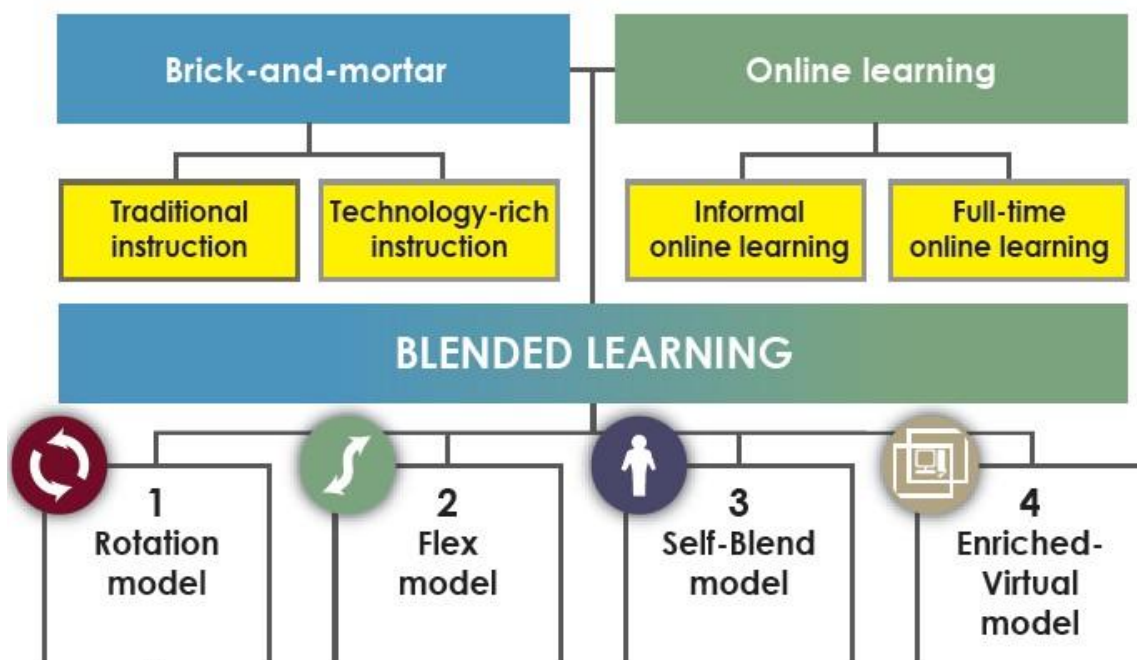
Characteristics of E-Learning

- Cost effective
- Improved performance and productivity
- Convenience and flexibility
- Timely feedback
- Easier access to information
- Improved retention

Blended Learning:

To accomplish any difficult educational goal, we often **must blend different media, instructional strategies, and design approaches**. This process of crafting custom mixes of learning goes under the name *blending*.

Blending can be defined simply as the mixture of different forms of education or training for a single purpose. The only thing remarkable about this definition is that it does not limit itself to mixtures of elearning and classroom learning. **A blend can be any mixture of any form of learning possible: classroom, virtual-classroom, or standalone e-learning.**



Blended learning has a strong dependence on the technical resources or tools with which the blended learning experience is delivered.

Synchronous and Asynchronous Learning

E-learning may either be synchronous or asynchronous. Synchronous learning occurs in real-time, with all participants interacting at the same time, while asynchronous learning is self-paced and allows participants to engage in the exchange of ideas or information without the dependency of other participants' involvement at the same time.

Synchronous learning:

Synchronous learning refers to the exchange of ideas and information with one or more participants during the same period. Examples are face-to-face discussion, online real-time live teacher instruction and feedback, Skype conversations, and chat rooms or virtual classrooms where everyone is online and working collaboratively at the same time.

Asynchronous:

Asynchronous learning may use technologies such as email, blogs, wikis, and discussion boards, as well as web-supported textbooks, hypertext documents, audio, video courses, and social networking using web 2.0. At the professional educational level, training may include virtual operating rooms.

<i>Synchronous vs. asynchronous e-Learning</i>		
	Distinctive features	Examples
Synchronous e-Learning	• Real-time • Live • Usually scheduled and time-specific (but can be impromptu) • Collective and often collaborative • Simultaneous virtual presence (with other learners and facilitators or instructors) • Concurrent learning with others	• Instant messaging • Online chat • Live Webcasting • Audioconferencing • Videoconferencing • Web conferencing
Asynchronous e-Learning	• Intermittent access or interaction • Self-paced • Individual, or intermittently collaborative • Independent learning • Usually available any time • Recorded or pre-produced	• E-mail • Threaded discussion • Boards • Web-based training • Podcasting • DVD • Computer-based training

E-learning tools

- Mobile computer technology
- Graphical multimedia
- Learning management System
- Computerized aided instruction
- Different types of computerized and Networking devices.

Technology

- Online internet based
offline
- Communication TOOLS
- Communication component (Sender, Receiver, Messages, protocol, Media)
- Communication tools
- Networking Devices
- Protocol
- Communication media Internet/ any
medium.
- Content
- Learner

Different Social Networking Side tools:

Concept of E-Learning 3.0

The concept of E-Learning 3.0 represents an evolution beyond its predecessors, E-Learning 1.0 and E-Learning 2.0. Let's delve into what this next stage entails:

E-Learning 1.0: This initial phase emerged around a decade ago. It started as a radical idea, with its effectiveness yet to be proven. Over time, it matured into a widely accepted mainstream approach. E-Learning 1.0 primarily focused on delivering educational content through digital platforms.

E-Learning 2.0: The shift to E-Learning 2.0 was driven by the emergence of Web 2.0 technologies. Social learning became a key aspect, emphasizing collaboration and the use of tools like blogs, wikis, podcasts, and virtual worlds. E-Learning 2.0 successfully integrated social software into the learning experience.

E-Learning 3.0: Now, we stand on the verge of another transformation. E-Learning 3.0 is poised to leverage the technologies that will shape Web 3.0. Here are some key aspects: **Artificial Intelligence (AI):** AI, particularly machine learning and data mining, will play a pivotal role in E-Learning 3.0. Imagine machine-understandable educational materials that adapt and respond to individual learners. AI will enhance personalization, adaptability, and responsiveness.

Web 3.0 Technologies: E-Learning 3.0 will benefit from advancements such as cloud computing, collaborative intelligent filtering, reliable data storage, higher screen resolutions, multi-gesture devices, and 3D touch interfaces. These technologies will empower the next generation of e-Learning. Web 2.0 describes World Wide Web sites that emphasize user-generated content, usability, and interoperability. The term was popularized by Tim O'Reilly and Dale Dougherty at the O'Reilly Media Web 2.0 Conference in late 2004, though it was coined by Darcy DiNucci in 1999. A Web 2.0 site may allow users to interact and collaborate with each other in a social media dialogue as creators of

user-generated content in a virtual community, in contrast to Web sites where people are limited to the passive viewing of content.

The key features of Web 2.0 include:

- **Folksonomy** - free classification of information; allows users to collectively classify and find information (e.g. tagging)
- **Rich User Experience** - dynamic content; responsive to user input
- **User Participation** - information flows two ways between site owner and site user by means of evaluation, review, and commenting. Site users add content for others to see
 - **Software as a service** - Web 2.0 sites developed APIs(Application Programming Interface) to allow automated usage, such as by an app or mashup.
 - **Mass Participation** - Universal web access leads to differentiation of concerns from the traditional internet user base.
 - Now after discussion about web 1.0 and 2.0 let's discuss on the propagation VS of web 2.0 i.e. is Web 3.0.
 - **Web 3.0 and AI:** Now, we're on the cusp of another transformation. Just as ELearning 2.0 replaced its predecessor, we anticipate the emergence of E-Learning 3.0. The driving force behind this evolution lies in the technologies that will shape Web 3.0. Web 3.0 is still in its infancy, but early indicators suggest that Artificial Intelligence (AI) will play a pivotal role. Specifically, machine learning and data mining are poised to be integral components of the Web 3.0 landscape.

E-Learning 3.0 Characteristics:

- **Anyone, Anywhere, anytime:** E-Learning 3.0 will continue to break down barriers. Learners will have access to educational content regardless of their location or time zone.
- **AI-Driven Adaptability:** Machine-understandable educational material will form the foundation. AI will interpret and utilize information to benefit authors and educators, making E-Learning 3.0 platforms highly adaptable and responsive to individual learners

Technological Enablers on e-learning 3.0:

Cloud Computing: Scalable and flexible infrastructure for seamless learning experiences.

Collaborative Intelligent Filtering: Personalized content recommendations based on learner preferences.

Increased Data Storage Capacity: Handling vast amounts of educational data.

Higher Screen Resolutions: Enhanced visual experiences.

Multi-Gesture Devices and 3D Touch Interfaces: Interactive learning interfaces

Concept of virtual learning environment

- A virtual learning environment (VLE) is a collaborative platform that enables better learning delivery¹. VLE learning platforms commonly allow²:
- **Content management** – creation, storage, access to and use of learning resources
- **Curriculum mapping and planning** – lesson planning, assessment and personalisation of the learning experience

- **Learner engagement and administration** – managed access to learner information, resources, and tracking of progress and achievement
- **Communication and collaboration** – emails, notices, chat, wikis, blogs



Digital Pedagogy and the TPACK Framework

- Digital pedagogy is an approach to teaching and learning that leverages digital technologies to enhance the educational experience. It involves thoughtfully integrating digital tools and resources into the curriculum, fostering student engagement, and promoting critical thinking and creativity.
- The TPACK framework (Technological Pedagogical Content Knowledge) is a model that provides a structure for understanding the complex interplay between technology, pedagogy, and content knowledge in effective teaching. It highlights the importance of teachers possessing a deep understanding of all three elements and how they intersect.

Key Components of TPACK

Technological Knowledge (TK): Understanding of digital tools and resources, their capabilities, and limitations. **Pedagogical Knowledge (PK):** Knowledge of teaching and learning principles, instructional strategies, and assessment methods. **Content Knowledge (CK):** In-depth understanding of the subject matter being taught.

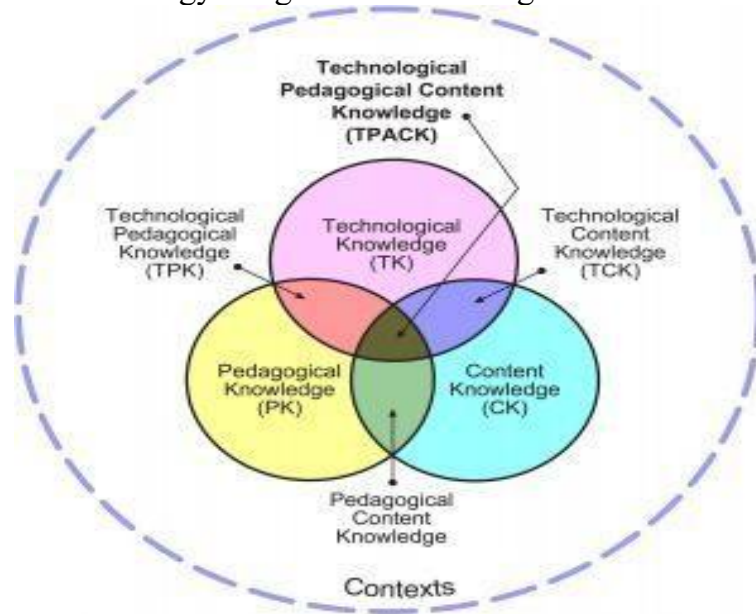
Intersections of TPACK

Technological Pedagogical Knowledge (TPK): Understanding how to use technology effectively for teaching and learning.

Technological Content Knowledge (TCK): Knowledge of how technology can be used to represent, transform, and personalize content.

Pedagogical Content Knowledge (PCK): Understanding how to teach a particular subject effectively, including common misconceptions and effective instructional strategies.

Technological Pedagogical Content Knowledge (TPACK): The intersection of all three areas, representing the ideal state of technology integration in teaching.



Importance and challenges of E-Learning

Some of the challenges of e-learning include:

- Technical issues, such as unreliable internet connection, software glitches, or hardware problems.
- Lack of motivation, engagement, and interaction with instructors and peers, which can affect the learning outcomes and satisfaction.
- Time management and self-discipline skills, which are essential for online learners to balance their studies with other responsibilities and distractions.
- Limited feedback, communication, and socialization opportunities, which can cause isolation, confusion, and reduced skill development.
- Decreased knowledge retention and intellectual abilities, which can result from passive or superficial learning, lack of practice, or visual overload.

Importance of e-learning

- E-learning makes giving and receiving information faster, more prolific, and more profitable.
- E-learning is flexible and accessible.
- E-learning provides personalized learning experiences.
- E-learning is cost-effective.
- E-learning has a global reach.